



Complete Nmap Guide

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- **What is Nmap?**

Nmap (Network Mapper) is a free and open-source tool used for network discovery and security auditing.

- **Why do we use it?**

We use Nmap to scan networks, discover devices, detect open ports, identify services, and find vulnerabilities.

1. **Target Specification**

Description	Command	Example
Single IP Scan	<code>nmap <target></code>	<code>nmap 192.168.1.1</code>
Specific IPs Scan	<code>nmap <target1> <target2></code>	<code>nmap 192.168.1.1 192.168.1.2</code>
Range Scan	<code>nmap <range></code>	<code>nmap 192.168.1.1-254</code>
Domain Scan	<code>nmap <domain></code>	<code>nmap scanme.nmap.org</code>
CIDR Scan	<code>nmap <CIDR></code>	<code>nmap 192.168.1.0/24</code>
Scan from File	<code>nmap -iL <file></code>	<code>nmap -iL targets.txt</code>
Random Hosts Scan	<code>nmap -iR <count></code>	<code>nmap -iR 100</code>
Exclude Hosts	<code>nmap --exclude <target></code>	<code>nmap --exclude 192.168.1.1</code>

2. Scan Types

Type	Command	Example
TCP SYN Scan (default)	<code>nmap -sS <target></code>	<code>nmap -sS 192.168.1.1</code>
TCP Connect Scan	<code>nmap -sT <target></code>	<code>nmap -sT 192.168.1.1</code>
UDP Scan	<code>nmap -sU <target></code>	<code>nmap -sU 192.168.1.1</code>
Stealth Scans	<code>nmap -sF</code>	<code>-sX</code>

3. Host Discovery

Discovery Type	Command	Example
Ping Scan	<code>nmap -sn <target></code>	<code>nmap -sn 192.168.1.1</code>
List Targets Only	<code>nmap -sL <target></code>	<code>nmap -sL 192.168.1.1-3</code>
TCP ACK Scan	<code>nmap -PA<ports> <target></code>	<code>nmap -PA22,80 192.168.1.1</code>

4. Port Scanning

Type	Command	Example
Single Port Scan	<code>nmap -p <port> <target></code>	<code>nmap -p 80 192.168.1.1</code>
Port Range Scan	<code>nmap -p <range> <target></code>	<code>nmap -p 21-100 192.168.1.1</code>
All Ports Scan	<code>nmap -p- <target></code>	<code>nmap -p- 192.168.1.1</code>
Fast Scan (top 100 ports)	<code>nmap -F <target></code>	<code>nmap -F 192.168.1.1</code>

5. Service & Version Detection

Detection Type	Command	Example
Service Version Detection	<code>nmap -sV <target></code>	<code>nmap -sV 192.168.1.1</code>
Light Mode	<code>nmap -sV --version-light <target></code>	<code>nmap -sV --version-light 192.168.1.1</code>
Aggressive Mode	<code>nmap -sV --version-all <target></code>	<code>nmap -sV --version-all 192.168.1.1</code>

6. OS Detection

Type	Command	Example
OS Detection	<code>nmap -O <target></code>	<code>nmap -O 192.168.1.1</code>
Aggressive Guessing	<code>nmap -O --osscan-guess <target></code>	<code>nmap -O --osscan-guess 192.168.1.1</code>

7. Output Options

Type	Command	Example
Normal Output	<code>nmap -oN <file> <target></code>	<code>nmap -oN normal.txt 192.168.1.1</code>
XML Output	<code>nmap -oX <file> <target></code>	<code>nmap -oX xml.txt 192.168.1.1</code>
Show Open Ports Only	<code>nmap --open <target></code>	<code>nmap --open 192.168.1.1</code>

8. Timing & Performance

Feature	Command	Example
Speed Profiles (0-5)	<code>nmap -T<0-5> <target></code>	<code>nmap -T4 192.168.1.1</code>
Rate Limit	<code>nmap --max-rate <rate> <target></code>	<code>nmap --max-rate 100 192.168.1.1</code>

9. Firewall Evasion Techniques

Technique	Command
Adjust Timing	<code>nmap -T2 -Pn 192.168.1.1</code>
Packet Fragmentation	<code>nmap -f 192.168.1.1</code>
Custom MTU Size	<code>nmap --mtu 16 192.168.1.1</code>
Randomized Host Order	<code>nmap --randomize-hosts -iL targets.txt</code>
Decoys	<code>nmap -D RND:10,ME 192.168.1.1</code>
Spoof Source IP	<code>nmap -S 1.2.3.4 192.168.1.1</code>
MAC Address Spoofing	<code>nmap --spooof-mac 00:11:22:33:44:55 192.168.1.1</code>
Source Port Spoofing	<code>nmap --source-port 53 192.168.1.1</code>
Custom TTL	<code>nmap --ttl 128 192.168.1.1</code>
Proxy Chains	<code>nmap --proxies proxylist.txt 192.168.1.1</code>
Uncommon Protocols	<code>nmap -PE 192.168.1.1</code>
Random Payloads	<code>nmap --data-length 50 192.168.1.1</code>
DNS Enumeration	<code>nmap -sL 192.168.1.1/24</code>
Combined Techniques	<code>nmap -f -T2 -D RND:5 --spooof-mac o --source-port 443 192.168.1.1</code>

10. Nmap Scripting Engine (NSE)

Script Name	Example Command
<code>dns-brute</code>	<code>nmap -p 80 --script=dns-brute [target]</code>

http-enum	<code>nmap --script=http-enum testhtml5.vulnweb.com</code>
ssh-brute	<code>nmap -p 22 --script=ssh-brute --script-args userdb=users.txt,passdb=passwords.txt [target]</code>
smb-enum-shares	<code>nmap --script=smb-enum-shares --script-args smbuser=guest,smbpass=guest [target]</code>
mysql-brute	<code>nmap -p 3306 --script=mysql-brute --script-args userdb=users.txt,passdb=passwords.txt [target]</code>
http-grep	<code>nmap -p 80 --script=http-grep --script-args http-grep.url=<subpage></code>
http-config-backup	<code>nmap -p 80 --script=http-config-backup</code>
smb-enum-users	<code>nmap --script=smb-enum-users [target]</code>
http-wordpress-enum	<code>nmap -p 80 --script=http-wordpress-enum [target]</code>
firewalk	<code>nmap --script=firewalk [target]</code>
mysql-empty-password	<code>nmap -p 3306 --script=mysql-empty-password [target]</code>
mysql-users	<code>nmap -p 3306 --script=mysql-users --script-args mysqluser=root,mysqlpass= [target]</code>
smb-os-discovery	<code>nmap -p 445 --script=smb-os-discovery [target]</code>
dns-zone-transfer	<code>nmap --script=dns-zone-transfer.nse --script-args dns-zone-transfer.domain=[domain] [target]</code>
ftp-anon	<code>nmap --script=ftp-anon [target]</code>
smtp-enum-users	<code>nmap --script=smtp-enum-users --script-args smtp.domain=[domain] [target]</code>
vulners	<code>nmap --script=vulners --script-args mincvss=[value] [target]</code>

snmp-brute	nmap --script=snmp-brute [target]
http-vuln-*	nmap --script=http-vuln-* [target]
http-title	nmap -p 80,443 --script=http-title [target-ip-or-domain]
ssl-cert	nmap -p 443 --script=ssl-cert [target-ip-or-domain]
vuln	nmap -p 80,443 --script=vuln [target-ip-or-domain]
http-robots.txt	nmap -p 80,443 --script=http-robots.txt [target-ip-or-domain]
ssh-hostkey	nmap -p 22 --script=ssh-hostkey [target-ip-or-domain]